

Anatomical structures detection using topological constraint knowledge in fetal ultrasound

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ABSTRACT

The accurate recognition of anatomical structures in fetal ultrasound images is crucial for prenatal diagnosis and determining ultrasound standard planes. However, this task can be arduous and time-consuming for sonographers. Existing methods usually treat anatomical structures as isolated individuals, ignore their interconnections, and lead to false and missed detections due to the similarities between these structures. In this paper, we propose a key anatomical structure-aware module, named KASM, which consists of class-aware representation learning and topological information interaction and is designed to integrate seamlessly into any proposal feature-based detection framework. Given the potential issue of information interactions between proposal features leading to feature over-smoothing, our approach adopts class-aware representation learning as an initial step. This process aims to enhance the intra-class compactness of proposal features while simultaneously dispersing inter-class features. The objective is to derive highly discriminative proposal features, thus facilitating subsequent accurate information interactions. Subsequently, proposal features are modeled using topological constraint relationships between anatomical structures and informative interactions are performed for structure recognition. We conduct extensive experiments on collected fetal ultrasound datasets in femoral, abdominal, and thalamic planes. The results consistently demonstrate that KASM effectively mitigates issues such as false detection and missed detection of anatomical structures across different baseline models, e.g., integrating KASM in Faster R-CNN reduces the false detection rate from 49.7% to 38.4% (−11.3%) and the missed detection rate from 60.8% to 57.0% (−3.8%) under the confidence threshold of 0.2. This underscores the broad applicability and efficacy of KASM in addressing these challenges.

1. Introduction

Ultrasound is widely used in fetal prenatal examination [1,2] and adult disease diagnosis [3,4] due to its low cost, high safety, and portability. It is an incredibly efficient tool for preventing birth defects and ensuring the well-being of individuals [5]. Among them, the recognition of key anatomical structures (KASs) in fetal ultrasound images is a common and practical task in prenatal ultrasonography [6,7]. This task involves localizing the position of KASs and classifying their categories, which aims to determine whether the ultrasound plane is standard or not by observing the presence or absence of KASs in the ultrasound plane.

In order to improve the efficiency and accuracy of prenatal ultrasonography, more and more researchers apply deep learning to

the intelligent analysis of ultrasound images [8–11]. A direct and simple way to deal with the task of automatic recognition of KASs in ultrasound images is to treat each structure as an isolated object and apply a deep neural network (DNN) to localize and classify each object. However, this way tends to yield false detection and missed detection cases. Fig. 1 illustrates this fact. False detection cases are incorrect identification of the categories of objects and even detection of anatomical structures that do not belong to a certain plane in a given image. The missed detection cases occur when the low confidence in an object causes it to be ignored. Nevertheless, these wrong results have relatively accurate localization results. Therefore, this fact indicates that the features extracted by the DNN in fetal ultrasound images can perform the task of localization of KASs relatively well.

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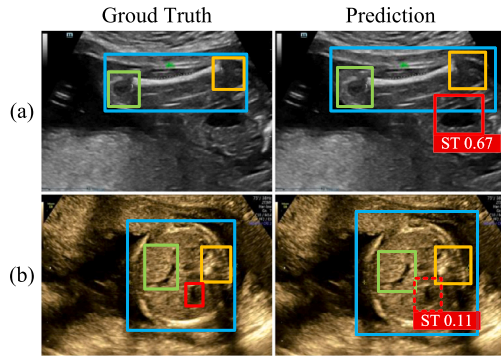


Fig. 1. Illustration of false and missed detection. (a) illustrates the detection of anatomical structures in the abdominal plane within the femoral plane with high confidence, leading to false detection, and (b) illustrates that anatomical structures detected in the abdominal plane are ignored due to low confidence, which is missed detection.

Unfortunately, due to the high similarity of features among KASs, as well as between KASs and the background, as illustrated in Fig. 3, successfully categorizing KASs becomes challenging.

In fetal ultrasound images, the appearance of KASs frequently exhibits topologically constrained relationships, such as coexistence, dependence, or exclusion. These relationships are often leveraged by sonographers to infer the presence or absence of another structure based on the known existing KASs [1,12,13]. Utilizing such relationships in DNN will hopefully reduce false and missed detection cases, but most previous methods ignore such relationships between KASs.

Therefore, for the task of automatic KAS recognition in fetal ultrasound images, it is crucial to capture and explore the topological constraints between KASs. This will help to strengthen the connection between KASs and improve the model recognition ability, which is important for improving the DNN classification accuracy. Currently, the popular transform module is highly regarded for its powerful long-range dependency modeling capability [14–16], but they consider the connections between image patches and still ignore the topological constraint relationships between KASs.

In this paper, we propose a novel module called Key Anatomical Structures-aware Module (KASM) that is specifically designed to model the relationships between KASs on a proposal feature-based object detection framework. Specifically, KASM consists of two steps: class-aware representation learning (CRL) and feature topological information interactions (TII). To address the challenges posed by high similarity between features and mitigate the risk of over-smoothing of features due to topological interactions, we propose CRL. This strategy aims to consolidate intra-class features while dispersing inter-class features to obtain feature representations that can distinguish between different KASs. This feature representation helps the DNN accurately recognize categories and facilitate subsequent information interactions. Note that this step is performed only during training. Then, in order to make full use of the inherent topological constraint relations and strengthen the connection between KASs, we propose the TII strategy. This strategy constructs a structural correlation matrix (SCM) based on a priori knowledge to guide topological information interaction for features with highly distinguishable categories to enhance feature representation. Therefore, our KASM encourages the model to construct feature representations that are highly distinguishable across categories and fully account for the presence of other KASs during the recognition process of each specific KAS. The method is conceptualized as shown in Fig. 2.

The main contributions of this paper are as follows.

- This paper introduces the KASM, a method for capturing topologically constrained relationships between KASs to strengthen structural associations, which can be integrated into any proposal feature-based detection framework.

- The cooperation of CRL and TII in KASM avoids the over-smoothing of features due to information interaction and effectively improves the problems of false and missed detections arising from highly similar features and isolated structures.
- We conduct extensive experiments on the data collected from three planes of the abdomen, thalamus, and femur. The experimental results consistently show that integrating KASM under different baselines can improve the model recognition ability and thus obtain more reasonable and accurate recognition results.

2. Related work

With the rapid development of deep learning techniques, DNN achieve impressive performance in image classification, object detection, and object segmentation, but there is still a lot of work devoted to extending the network to be more applicable to medical images.

Object detection. Object detection is the task of locating and classifying objects in images using DNN, which is commonly used in medical images for capturing anatomical structures [17–19]. Zhen et al. [20] detected anatomical structures in early pregnancy planes to locate standard planes based on the YOLOv3 network. Wu et al. [21] applied an AI recognition model to identify standard planes of the fetal heart, which compared to manual scanning, significantly saves time and ensures efficiency. Komatsu et al. [22] used DNN to detect abnormal structures in the fetal ultrasound video of the heart. Lin et al. [23] based on Faster R-CNN to detect KASs and ROIs in the fetal head planes to determine the standard planes.

Contrastive learning. Contrastive learning is commonly used to maximize consistency between similar instances, which are defined as pairs of positive samples, and to encourage differences between different instances or negative samples [24–26]. In [27], Chen et al. proposed an ultrasound contrastive learning framework, called USCL, which achieves superior performance over ImageNet pre-training by learning compact semantic clusters from ultrasound videos. In [28], Zhang et al. proposed a hierarchical contrastive learning (HiCo) method, which consists of pairwise hierarchical semantic alignment and cross-level semantic alignment, and can effectively improve the migrability of pre-training of ultrasound video analytics models. Sun et al. [29] addressed the issue of insufficiently representative features for distinguishing between benign and malignant nodules by employing contrastive learning, which can effectively extract the representative features and achieve the classification of thyroid nodules. Chartsias et al. [30] applied contrastive learning to echocardiographic view classification and showed that contrastive learning can effectively solve the data annotation bottleneck without degrading the overall performance. Fu et al. [31] used contrastive learning on positive/negative sample pairs of anatomical structures in ultrasound images to improve the visual representation of medical images.

Graph convolution networks. Each node in the graph convolution network (GCN) iteratively aggregates feature vectors from its neighborhood through message passing to compute new feature vectors for the next hidden layer in the network [32–34]. Ju et al. detailed the challenges faced by graph neural network models in four areas: imbalance, noise, privacy, and out-of-distribution [35], and demonstrated the superiority of the graph structure in zero-sample node classification [36]. Chen et al. [37] designed a deep learning model using graph convolution, which was able to perform the task of automatic segmentation and measurement of fetal lungs and heart regions. The extensive study of graph structure [38–40] has also inspired researchers to explore the field of medical imaging. Yin et al. [41] further optimized the instance-level features by exploring potential correlations between instances using GCN based on instance features extracted by a convolutional neural network. Zhao et al. [42] used graph structure learning hybrid knowledge graph to achieve the detection of standard planes of the fetal ultrasound head. Wang et al. [43] enhanced the ability of the network to perceive the boundary region by adding a GCN module, which effectively suppressed the interference of ultrasound image quality defects on fetal head circumference measurement.

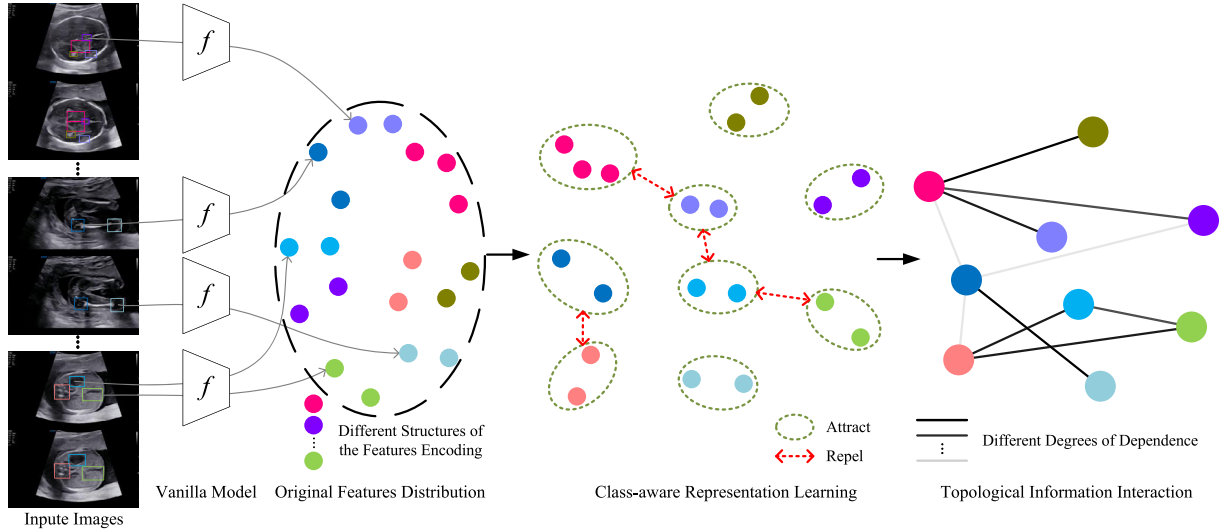


Fig. 2. Method conceptualization. We first introduce CRL in the original feature distribution to construct discriminative proposal features, and then perform TII among proposal features to fuse the features. Note that the introduction of CRL avoids the over-smoothing of features in TII.

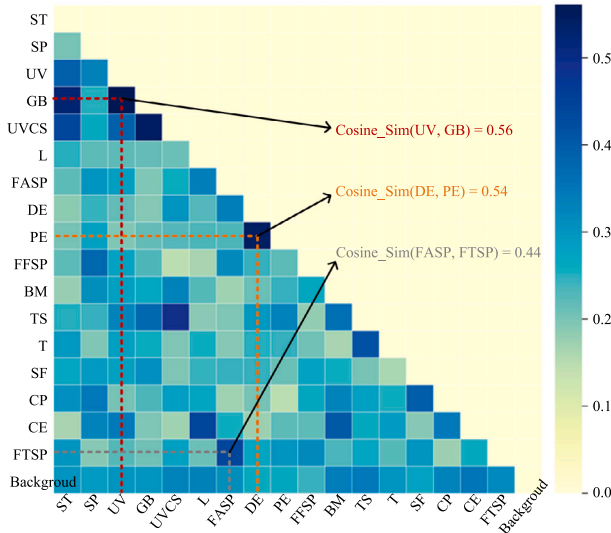


Fig. 3. Illustrates the cosine similarity between proposal features for different KASs. For example, the red dashed line corresponds to a result indicating that the similarity of the proposal features for the umbilical vein (UV) and gallbladder (GB) is 0.56.

3. Method

The overall structure of the integrated KASM is shown in Fig. 4. The proposal feature-based detection network ultimately feeds a set of proposal features $F \in \mathbb{R}^{N \times D}$ into the classification branch and regression branch respectively to obtain prediction results, where N is the number of features and D is the feature dimensions. However, this is unable to fully utilize the topological constraint relations inherent in ultrasound images. To capture the topological constraint relationships between structures, KASM constructs a feature-adaptive correlation matrix by predicting the degree of dependency between proposal features F . The process uses an SCM derived from prior knowledge statistics for supervised training. Subsequently, this matrix is employed to establish a graph pattern guiding the interaction of feature information. However, the interaction among proposal features with high similarity can result in over-smoothing, which undermines model recognition. Therefore, KASM employs CRL as an initial step to promote the closeness of features within the same category while

dispersing features across different categories. This ensures the creation of proposal features that exhibit high discriminative power within their respective categories.

3.1. Class-aware representation learning

The high similarity in features among KASs and between KASs and the background presents a challenge for the backbone network to generate highly discriminative proposal features. Moreover, TII may make similar features smoother, which can add additional stress to the classifier. To address this issue and learn proposal features from images that are more discriminative to classes, we propose to use CRL aimed at learning intra-class compactness and inter-class exclusion between proposal features.

Specifically, we introduce a projection layer that is parallel to the classification branch. In general, the proposal features are activated using ReLU [44] resulting in some of the features being truncated to zero, and thus the similarity between the proposal features cannot be computed directly [25]. Meanwhile, to reduce the computational effort, we use the projection layer to encode proposal features F as feature vector $f \in \mathbb{R}^{N \times d}$ with default $d = 128$. Thus, we can construct a set for each proposal feature $S_i = (F_i, f_i, u_i, y_i)$, where $F_i \in \mathbb{R}^{1 \times D}$ and $f_i \in \mathbb{R}^{1 \times d}$ represent a single proposal feature as well as the corresponding projection feature, respectively, $u_i \in \mathbb{R}^+$ denotes the intersection over union of the proposal feature box with the corresponding ground truth box, and y_i denotes the category label. Finally, we compute the cosine similarity between the projection features and improve the differentiation between the features by minimizing the InfoNCE loss [45]:

$$f_i = Proj(F_i), \quad (1)$$

$$Sim(F_i, F_j) = \frac{f_i^T \cdot f_j}{\|f_i\| \cdot \|f_j\|}, \quad (2)$$

$$L_{nce}(F_i) = \frac{-1}{N_{y_i}} \sum_{j=1, j \neq i}^N \mathbb{1}_{y_i=y_j} \log \frac{\exp(Sim(F_i, F_j))}{\sum_{k=1}^N \mathbb{1}_{k \neq i} \exp(Sim(F_i, F_k))}, \quad (3)$$

where $Proj$ denotes a projection layer consisting of a layer of fully connected layers, and Sim is a cosine similarity computation formula. To avoid learning of a bad proposal feature, i.e., the proposal feature box has a small coverage area of the ground truth box. We use u_i as the loss weight of the corresponding feature F_i to minimize the error, so that the final overall loss is calculated as follows:

$$L_{crl} = \frac{1}{N} \sum_{i=1}^N u_i \cdot L_{nce}(F_i). \quad (4)$$

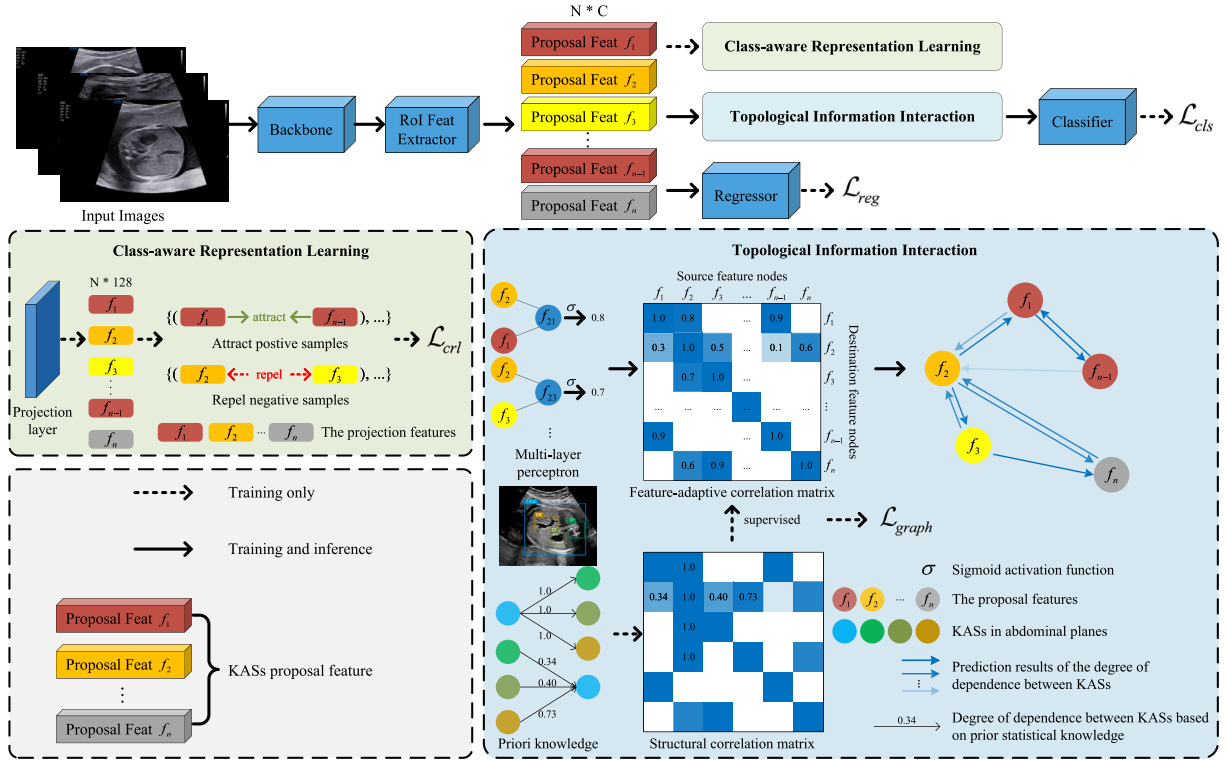


Fig. 4. The KASM contains two steps: CRL and TII. The first step CRL is performed only at training time and is used to provide proposal features that are highly discriminative to the category. The second step TII constructs a feature-adaptive correlation matrix for the proposal features and establishes a graph pattern based on this matrix to guide the information interaction between the features and enhance the feature representation. Note that the matrix is subject to supervised learning of structural correlation matrices based on a priori knowledge statistics.

3.2. Topological information interaction

Using the topological constraints inherent in ultrasound images, combining multiple KASs can accurately determine whether another KAS is present. This approach helps to reduce the probability of false and missed detection. We propose to construct a feature-adaptive correlation matrix by predicting correlations between features and constructing graph patterns based on this matrix for information interaction. Meanwhile, we use a statistical approach in the dataset to construct an SCM that is able to construct truth values by the category labels to supervise the learning of the feature-adaptive correlation matrix during the training process.

3.2.1. Structural correlation matrix

As illustrated in Fig. 5, in the fetal abdominal ultrasound image, the presence of the fetal abdomen standard plane (FASP) indicates the definite presence of a spine (SP). However, in the instance of the SP being present, it does not necessarily indicate the FASP. Hence, we model structural correlation based on the training dataset in the form of conditional probability, i.e., $P(\text{SP}|\text{FASP})$ denotes the probability of label SP occurring when label FASP appears. It is important to note that the SCM is asymmetric.

In particular, we calculate the frequency of co-occurrences between structures in the training set, along with the individual occurrence count of each structure. Subsequently, we construct the SCM by dividing the corresponding elements of the two matrices, i.e., $X \in \mathbb{R}^{c \times c}$, $Y \in \mathbb{R}^{c \times 1}$ and $Z = X/Y \in \mathbb{R}^{c \times c}$, where c is the number of categories, X_{ij} denotes the number of co-occurrences of structure i and structure j , Y_i denotes the number of occurrences of structure i , and Z_{ij} denotes the degree of dependency of structure j on structure i . Meanwhile, to avoid the influence of noise contained in the data labeling information, we use a threshold θ to further filter the matrix,

$$P_{ij} = \begin{cases} 0 & \text{if } Z_{ij} < \theta, \\ Z_{ij} & \text{otherwise,} \end{cases} \quad (5)$$

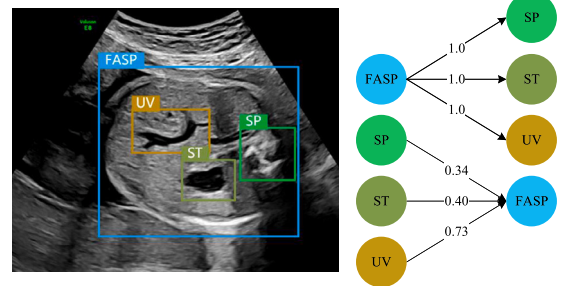


Fig. 5. Illustration of conditional probability in the FASP. While the appearance of FASP implies the presence of SP, stomach, and UV, it does not guarantee that FASP will present when SP is present.

where $P \in \mathbb{R}^{c \times c}$ represents the SCM obtained from the statistical data analysis, and θ is set to 0.1. Note that we define the degree of dependency between the structure and the background as zero.

3.2.2. Feature-adaptive correlation matrix

We construct a feature-adaptive correlation matrix for the N proposal features F to perform topological interactions between feature nodes in the graph pattern, and utilize the SCM obtained via statistics in Section 3.2.1 for supervised training.

Specifically, we construct the graph $G = (N, E)$ with N proposal features as nodes in the graph and predict correlations between nodes as weights of edges using a multi-layer perceptron (MLP). To further utilize the feature difference information among nodes, we jointly input the node features F_i , F_j , and $F_i - F_j$ into the MLP to obtain the prediction results:

$$W_{e_{ij}} = \sigma(\text{MLP}(F_i, F_j, F_i - F_j)), \quad (6)$$

where MLP consists of two fully connected layers and σ denotes sigmoid activation.

Eventually, we can obtain the feature-adaptive correlation matrix, i.e., $W_e \in \mathbb{R}^{N \times N}$, where $W_{e_{ij}}$ represents the predicted value of node j dependence on node i . We then use the SCM and the category label y_i matched by each proposal feature F_i to construct the learning objective $M \in \mathbb{R}^{N \times N}$ for the feature-adaptive correlation matrix, where $M_{ij} = P_{y_i, y_j}$ denotes the correlation label between proposal features F_i and F_j . The learning of the feature-adaptive correlation matrix can be bootstrapped by enforcing the predicted value of $W_{e_{ij}}$ to be equal to the ground truth M_{ij} , using a loss function L_{graph} :

$$L_{graph} = \sum_{i=1}^N \sum_{j=1}^N (W_{e_{ij}} - M_{ij})^2. \quad (7)$$

3.2.3. Information interaction

After obtaining the feature-adaptive correlation matrix, we can augment each feature with different degrees of dependency between nodes, which is computed as follows:

$$F'_i = W_1 F_i + W_2 \sum_{j=1}^N W_{e_{ij}} F_j, \quad (8)$$

where W_1 and W_2 are learnable matrix parameters, $W_{e_{ij}}$ is the feature-adaptive correlation matrix mentioned in Section 3.2.2, F_i and F_j are the features of the two nodes, respectively, and F'_i denotes the enhanced features after interaction and used for category recognition. The overall loss is calculated as follows:

$$L = L_{cls} + L_{reg} + L_{crl} + L_{graph}, \quad (9)$$

where L_{cls} and L_{reg} denote the category classification loss and the bounding box regression loss in the vanilla network respectively.

Algorithm 1 The inference process of the integrated KASM model.

Input:

$F \in \mathbb{R}^{N \times D}$ is the proposal feature generated by the vanilla model for an ultrasound image.

Output:

$F' \in \mathbb{R}^{N \times D}$ is the enhanced feature after the topological information interaction for F .

$C \in \mathbb{R}^{N \times c}$ is the classification result of the classifier for F' .

$B \in \mathbb{R}^{N \times 4}$ is the regression result of the regressor for F' .

- 1: Dependency prediction $W_e \in \mathbb{R}^{N \times N}$ between features F obtained through Eq. (6);
 - 2: Construct a graph pattern $G = (N, E)$, where the N proposal features F serve as the set of nodes and the dependencies W_e between nodes serve as the set of edges;
 - 3: The enhanced features F' are obtained by interacting the source features F according to Eq. (8) under the graph structure G ;
 - 4: $C \leftarrow \text{Classifier}(F')$;
 - 5: $B \leftarrow \text{Regressor}(F)$;
 - 6: **return** F' , C and B .
-

However, it is worth noting that since the regression of bounding boxes is not restricted by topological constraints and to obtain proposal features that are closer to the ground truth box, we utilize the feature F' for the category recognition task and the feature F for the bounding box regression task. The inference and training processes for the integrated KASM model are shown in Algorithms 1 and 2, respectively.

4. Experiments

4.1. Dataset

We obtain the approval of the Ethics Committee of Shenzhen Maternity and Child Healthcare Hospital to collect 18,266 fetal ultrasound

Algorithm 2 The training process of the integrated KASM model.

Input:

$F \in \mathbb{R}^{N \times D}$ is the proposal feature generated by the vanilla model for an ultrasound image.

Output:

$\mathcal{L}_{cls}$, $\mathcal{L}_{reg}$, $\mathcal{L}_{crl}$ and $\mathcal{L}_{graph}$ represent classification loss, regression loss, CRL loss, and modeling graph structure loss, respectively.

- 1: Same as Algorithm 1 to obtain the feature-adaptive correlation matrix W_e , classification result C and regression result B ;
 - 2: Calculate $\mathcal{L}_{cls}$ and $\mathcal{L}_{reg}$ from the loss function of the vanilla model; // Class-aware representation learning
 - 3: $f \in \mathbb{R}^{N \times 128} \leftarrow \text{Proj}(F)$;
 - 4: Calculate the feature similarity between f ;
 - 5: Calculate the loss $\mathcal{L}_{crl}$ from Eq. (4); // Topological Information Interaction
 - 6: Statistics the structural correlation matrix $P \in \mathbb{R}^{c \times c}$ in the training set;
 - 7: Construct the learning objective $M \in \mathbb{R}^{N \times N}$ of the W_e based on the P and the category labels of the proposal feature F matching;
 - 8: Calculate the loss $\mathcal{L}_{graph}$ from Eq. (7);
 - 9: **return** $\mathcal{L}_{cls}$, $\mathcal{L}_{reg}$, $\mathcal{L}_{crl}$ and $\mathcal{L}_{graph}$.
-

images in the abdominal, femoral, and thalamus planes on several ultrasound devices such as Samsung and Siemens. We randomly select 14,927 images and 3339 images as the training and validation sets for related experiments, respectively. 17 KASs in the three fetal ultrasound planes, as shown in Table 1, are manually labeled by sonographers with at least two years of experience.

4.2. Implementation details

All experiments are conducted on a Linux server with an NVIDIA RTX 3090 GPU. The proposed method is implemented using PyTorch 1.13.0 and Python programming language. In addition, the image size in the experiments is set to 608*608, and the parameter settings such as learning rate and optimizer are the same as the baseline used, and for the different comparison baselines, we uniformly train 35 epochs and set the batch size to 8.

For more intuitive analysis, we additionally use the false detection ratio (FDR) and the missed detection ratio (MDR) as metrics for assessing the model performance. For the detection results at a certain confidence threshold α , we denote the number of images with false detection cases as FD , the number of images with missed detection cases as MD , and the total number of images as T . Therefore, the FDR_α and the MDR_α at that confidence threshold α can be computed as: $\frac{FD}{T}$ and $\frac{MD}{T}$, respectively.

4.3. Results

To thoroughly evaluate the effectiveness of KASM, we first apply KASM on different baselines to demonstrate the module's general applicability in different architectures based on proposal boxes. Then, we compare three related approaches to demonstrate the effectiveness of KASM. Finally, we also perform ablation studies and feature visualization on KASM to demonstrate the necessity of CRL and TII.

4.3.1. Performance comparison on different baselines

In this subsection, we evaluate KASM in currently popular proposal features-based network architectures including Cascade R-CNN [46], Sparse R-CNN [47], and DiffusionDet [48].

Table 2 shows the experiment results before and after the integration of KASM. Notably, after integrating KASM, the accuracy of each network is significantly improved for KAS lung (L) and cerebellum (CE),

Table 1

Key anatomical structures are included in the three planes.

Planes	Id	Anatomic structures	Abbreviation	Training set	Testing set
Abdominal plane	0	Stomach	ST	4278	895
	1	Spine	SP	4985	1004
	2	Umbilical vein	UV	2334	585
	3	Gallbladder	GB	2497	381
	4	Umbilical vein cross-section	UVCS	2309	341
	5	Lung	L	1099	255
Femoral plane	6	Fetal abdomen standard plane	FASP	1690	394
	7	Distal epiphysis	DE	4972	1091
	8	Proximal epiphysis	PE	4951	1091
Thalamic plane	9	Fetal femur standard plane	FFSP	4916	1090
	10	Brain middle	BM	8260	2094
	11	Transparent septum	TS	4168	1100
	12	Thalamus	T	9118	2346
	13	Sylvian fissure	SF	4540	1141
	14	Choroid plexus	CP	4138	1044
	15	Cerebellum	CE	947	273
	16	Fetal thalamus standard plane	FTSP	3232	863

even with limited training data. This improvement can be attributed to KASM's modeling of topological relationships between KASs. Specifically, there exists an exclusionary relationship between L and FASP, as well as between CE and fetal thalamus standard plane (FTSP). KASM can readily suppress the occurrence of L and CE once FASP and FTSP have been clearly defined by incorporating other KASs, thus improving accuracy.

Fig. 7 shows the false and missed detection ratios of each baseline network before and after the integration of KASM at different confidence thresholds. A point of interest is that, due to the integration of multiple key anatomical structures by KASM, a higher confidence level is also assigned to the structure when a false detection situation occurs, and thus the false detection ratio of KASM may gradually approach or even exceed the false detection ratio of the vanilla network when the confidence threshold is gradually increased, but such an error is much smaller than the gain brought by the missed detection ratio, e.g., in Table 2, the integrated KASM of Sparse R-CNN improves the false detection ratio by 1.8% but reduces the missed detection ratio by 11.0% at a confidence threshold of 0.8.

Fig. 8 shows the detection results of the competing methods on the test dataset before and after using KASM. Compared to the original method, the method after integrating KASM scores higher on missed and correctly detected KASs.

4.3.2. Performance comparison with competitive methods

To further evaluate the performance of our method, we compare the effectiveness of KASM applied to various types of baselines with other related methods, including HKRM [49], USPD [42], and FCUM [11]. Among them, HKRM is a method that applies topological constraint relations to natural images, while USPD and FCUM focus on the detection of fetal ultrasound images.

Table 2 shows the experiment results of these three methods. Although these three methods show an advantage in the accuracy of some KASs, they still underperform for L and CE, which highlights the necessity of modeling the topological relationship between KASs and the effectiveness of KASM. Notably, FCUM achieves the lowest level of missed detection when the confidence threshold is 0.2. We believe that this is due to the advantages of the single-stage dense target detection scheme. Meanwhile, the low false detection rate of FCUM at a confidence threshold of 0.8 may be attributed to the general lack of confidence in the detection results, which is verified by the ultra-high missed detection rate and Fig. 8. Overall, the various types of baselines with KASM showed more stable and superior performance.

Fig. 8 illustrates the detection results of the relevant methods on the test dataset. All three methods have different degrees of false detection problems. For example, in the second detection result of HKRM, KASs in other planes are incorrectly detected in the femoral plane; in the fifth

detection result of USPD, both umbilical vein and umbilical vein cross-sections are detected; and in the third and fourth detection results of FCUM, the coexistence of FTSP and CE occurs. In contrast, the method using KASM shows superior performance in terms of accuracy and stability.

4.3.3. Ablation study

We take Faster R-CNN [50] as the baseline to analyze the impact of the components CRL and TII in KASM on the performance of the model, and the results are shown in Table 3. CRL and TII can bring consistent improvements to model performance. In short, the integration of KASM with the baseline Faster R-CNN model improves the $mAP_{50:95}$ metric from 51.1% to 55.6% (+4.5%) and reduces the false and missed detection ratio substantially.

Furthermore, we use t-SNE [51] to further visualize the distribution status of proposal features before and after employing CRL and TII, respectively. As shown in Fig. 6, (b) after CRL, the feature distribution exhibits more pronounced intra-class compactness and inter-class separateness, but each structural feature exists in isolation without establishing connections between structures. (c) after TII, the KAS features within the same plane type demonstrate apparent aggregation, but lack well-defined boundary distances between features. By simultaneously applying these two steps in (d), the visualization indicates a denser distribution of KAS features within the same plane type, with well-defined boundary distances between features. This illustrates the necessity and effectiveness of combining CRL in TII.

Finally, we borrow the metrics of node smoothness level (NSL) and graph smoothness level (GSL) from [52] to evaluate feature smoothing to further validate the effectiveness of CRL. Specifically, NSL and GSL are defined as follows:

$$NSL_i = \frac{1}{N-1} \sum_{j \in \mathcal{V}, j \neq i} \frac{X_i \cdot X_j}{|X_i| |X_j|} \quad (10)$$

$$GSL = \frac{1}{N} \sum_{i \in \mathcal{V}} NSL_i \quad (11)$$

where $\mathcal{V}$ represents the set of nodes in the graph and i and j represent the nodes respectively. NSL_i measures the average similarities between node i and every other node in the graph. Corresponding to NSL_i , GSL measures the average similarities between the node pairs in the graph. Note that both metrics are positively correlated to the smoothness level.

As shown in Table 4, we calculate the GSL for the proposal features before and after using CRL. The results show that CRL has a positive effect in mitigating the over-smoothing problem.

In conclusion, the ablation experiments on components in KASM, the feature distribution study, GSL analysis, and the visual analysis of the detection results further confirm the effectiveness of our module in solving the problem of KAS errors and missed detections.

Table 2

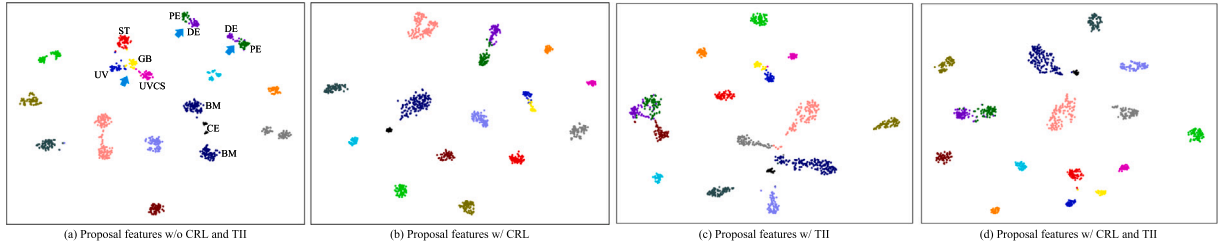
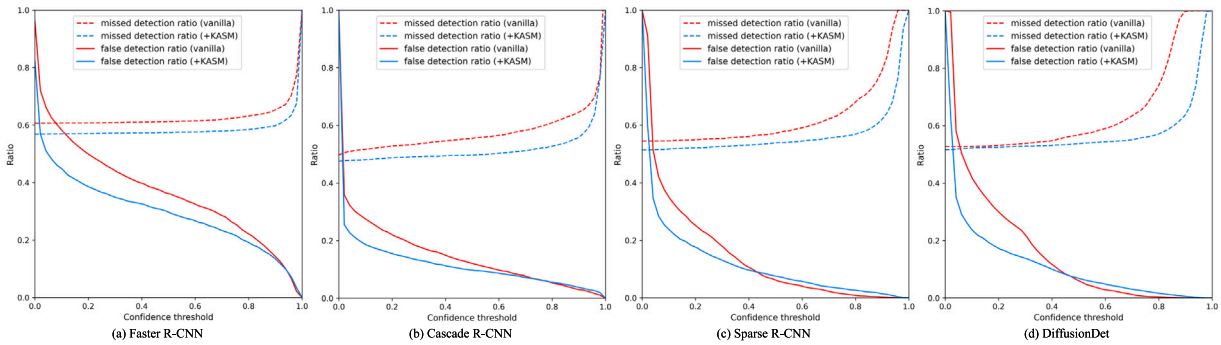
Detection results on the test dataset. KASM denotes the integration of our modules, including CRL and TII.

Methods	$AP_{50} \uparrow$																	$mAP_{50} \uparrow$	$mAP_{50:95} \uparrow$	$FDR_{0.2} \downarrow$	$MDR_{0.2} \downarrow$	$FDR_{0.8} \downarrow$	$MDR_{0.8} \downarrow$	$Params(M)$	FPS
	ST	SP	UV	GB	UVCS	L	FASP	DE	PE	FFSP	BM	TS	T	SF	CP	CE	FTSP								
Faster R-CNN	98.4	98.8	96.7	95.3	66.8	52.6	83.2	96.8	95.7	99.8	86.5	89.0	95.8	98.1	88.0	79.0	95.9	89.2	51.1	49.7	60.8	23.0	63.1	31.35	32
Faster R-CNN (+KASM)	98.5	98.8	96.5	97.2	75.6	61.5	86.4	97.0	97.0	99.9	91.6	91.6	96.6	98.5	90.9	82.1	95.8	91.5 (+2.3%)	55.6 (+4.5%)	38.4 (-11.3%)	57.0 (-3.8%)	19.7 (-3.0%)	58.4 (-4.7%)	31.53	27
Cascade R-CNN	98.3	99.4	97.1	96.0	88.1	56.0	89.8	98.2	96.9	99.5	97.1	98.4	97.7	99.4	98.9	80.6	97.5	93.4	66.0	21.8	52.9	5.4	60.8	70.80	27
Cascade R-CNN (+KASM)	98.3	99.4	97.2	96.9	89.5	58.0	91.1	98.3	98.1	99.4	97.9	98.5	98.6	99.4	98.9	81.5	98.2	94.0 (+0.6%)	66.6 (+0.6%)	15.4 (-6.4%)	48.8 (-4.1%)	5.6 (+0.2%)	52.7 (-8.1%)	71.00	23
Sparse R-CNN	98.6	98.6	95.0	96.4	84.6	57.3	90.4	97.3	96.8	98.8	97.0	98.7	96.7	98.8	97.5	79.3	98.5	93.0	65.6	24.9	55.0	0.9	67.7	106.25	25
Sparse R-CNN (+KASM)	98.7	98.7	95.6	96.2	85.1	58.0	90.3	97.7	97.2	98.9	97.1	98.7	96.8	98.8	97.6	82.9	98.4	93.3 (+0.3%)	66.2 (+0.6%)	17.4 (-7.5%)	52.1 (-2.9%)	2.7 (+1.8%)	56.7 (-11.0%)	106.49	24
DiffusionDet	98.7	98.9	95.3	96.2	85.7	52.4	88.9	98.0	97.2	98.9	97.1	97.7	97.2	98.8	96.6	80.2	97.6	92.7	65.5	29.7	53.2	0.4	73.3	110.70	22
DiffusionDet (+KASM)	98.6	98.9	94.6	96.5	87.5	56.9	89.5	97.6	97.2	98.9	97.3	98.6	97.3	98.7	97.7	82.4	98.2	93.3 (+0.6%)	66.7 (+1.2%)	17.0 (-2.7%)	52.4 (-0.8%)	1.6 (+1.2%)	56.9 (-16.4%)	110.93	21
Other methods																									
HKRM	98.2	98.8	96.2	96.5	79.4	57.3	75.3	97.4	96.8	99.9	89.7	90.6	96.5	98.5	90.4	73.2	89.3	89.6	53.7	20.6	53.6	10.1	60.9	31.69	26
USPD	98.6	98.7	96.8	97.8	81.1	57.5	72.9	97.2	96.3	98.9	92.8	91.8	96.3	98.3	90.4	66.0	81.8	89.0	51.2	34.5	48.6	14.2	60.7	33.63	12
FCUM	98.6	98.8	96.6	96.5	89.8	55.7	88.4	60.6	50.1	99.8	96.6	97.8	96.6	98.8	97.4	79.7	97.7	88.2	61.0	29.9	36.9	1.9	81.0	417.28	21

Table 3

Ablation studies for each component. CRL and TII are denoted as class-aware representation learning and topological information interaction, respectively.

Method	$AP_{50} \uparrow$																	$mAP_{50} \uparrow$	$mAP_{50:95} \uparrow$	$FDR_{0.2} \downarrow$	$MDR_{0.2} \downarrow$	$FDR_{0.8} \downarrow$	$MDR_{0.8} \downarrow$
	ST	SP	UV	GB	UVCS	L	FASP	DE	PE	FFSP	BM	TS	T	SF	CP	CE	FTSP						
Faster R-CNN	98.4	98.8	96.7	95.3	66.8	52.6	83.2	96.8	95.7	99.8	86.5	89.0	95.8	98.1	88.0	79.0	95.9	89.2	51.1	49.7	60.8	23.0	63.1
Faster R-CNN +CRL	98.4	98.9	96.9	96.4	74.3	60.7	89.2	96.6	96.6	99.0	87.9	88.9	96.5	98.3	87.9	80.3	97.2	90.8 (+1.6%)	54.0 (+2.9%)	41.4 (−8.3%)	58.8 (−2.0%)	22.6 (−0.4%)	60.5 (−2.6%)
Faster R-CNN +CRL +TII	98.5	98.8	96.5	97.2	75.6	61.5	86.4	97.0	97.0	99.9	91.6	91.6	96.6	98.5	90.9	82.1	95.8	91.5 (+2.3%)	55.6 (+4.5%)	38.4 (−11.3%)	57.0 (−3.8%)	19.3 (−3.7%)	58.4 (−4.7%)

**Fig. 6.** Results of t-SNE visualization of proposal features. CRL and TII are denoted as class-aware representation learning and topological information interaction, respectively.**Fig. 7.** The changes in false and missed detection ratios before and after the integration of KASM for each baseline network at different confidence thresholds.**Table 4**

Impact of components on GSL.

Method		$GSL \downarrow$
Faster R-CNN	+ TII	0.503
Faster R-CNN	+ TII + CRL	0.479

5. Discussion

The KASM involves CRL and TII, both steps that positively affect the detection results. In Fig. 6, the t-SNE visualization demonstrates

that the CRL brings the spatial distribution of features within the same category closer together, enhancing their representativeness and thereby contributing to the topological interaction among features. The detection results in Fig. 8 demonstrate that each network, after integrating KASM, is capable of effectively reducing the confidence of incorrect categories and increasing the confidence of correct categories, benefiting from the topological interactions between features.

Although the networks integrating KASM show better results in the recognition of KASs in fetal ultrasound images, there are still several issues that need to be explored in further study.

(1) The SCM used for supervised feature-adaptive correlation matrix training relies on statistics from the dataset and is constrained by the setting of hyperparameter θ . Therefore, the confidence of this matrix



Fig. 8. Visualization of detection results. The confidence threshold is 0.1 and the IOU threshold is 0.3.

is restricted when dealing with datasets containing a large number of noisy annotations or exhibiting severe long-tailed distributions.

(2) By utilizing topological constraint relations, such as the coexistence of KASs, KASM effectively overcomes the challenges of false and missed detection in category classification. However, as shown in the second and third rows of the second column in Fig. 8, the detection results that conform to the coexistence relationship but in the wrong

location cannot currently be addressed by KASM. Considering that most of the variations in KAS positions in different ultrasound images are due to the different rotation angles of the sonographer, we will consider two approaches to solve such problems in our future work. (i) We will learn the rotation angle of ultrasound images and consider it as a regression problem. The prediction results will be used to rotate the image to the canonical angle and then determine whether the relative positions

between the structures are correct. (ii) We will create canonical graph patterns for different types of planes and verify that the predicted results for that plane image conform to the pattern.

6. Conclusion

In this paper, we propose the KASM module that can be integrated into the proposal feature-based detection network to accomplish the task of recognizing KASs in fetal ultrasound images. To overcome the challenge of recognizing anatomical structure features, with their high similarity, and prevent excessive smoothing caused by topological interaction, we introduce CRL to increase the feature distances between different categories while minimizing the feature distances within the same category. To improve the problems of false and missed detection in the recognition of KASs, we propose the TII to model the correlation between KASs. This step builds a feature-adaptive correlation matrix between features and constructs a graph pattern based on this matrix for interaction to enhance the feature representation.

The experiment results show that KASM can be integrated into the proposal feature-based detection network to improve detection accuracy and effectively reduce false and missed detection problems. Our work rethinks the importance of latent knowledge in fetal ultrasound images for KAS recognition, demonstrating that exploring and mining latent knowledge is beneficial in improving the problems of false and missed detection, and is expected to be extended to other types of images.

CRediT authorship contribution statement

Juncheng Guo: Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Guanghua Tan:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Jianxin Lin:** Writing – review & editing, Methodology, Conceptualization. **Bin Pu:** Writing – review & editing, Validation, Investigation. **Xin Wen:** Validation, Investigation, Data curation. **Chunlian Wang:** Resources, Data curation. **Shengli Li:** Project administration, Data curation. **Kenli Li:** Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

References

- [1] L.J. Salomon, Z. Alfrevic, V. Berghella, C. Bilardo, E. Hernandez-Andrade, S. Johnsen, K. Kalache, K.-Y. Leung, G. Maling, H. Munoz, et al., Practice guidelines for performance of the routine mid-trimester fetal ultrasound scan, *Ultrasound Obstet. Gynecol.* 37 (1) (2011) 116–126.
- [2] A. Khalil, A. Sotiriadis, R. Chaoui, F. da Silva Costa, F. D'Antonio, P. Heath, C. Jones, G. Maling, A. Odibo, F. Prefumo, et al., ISUOG practice guidelines: The role of ultrasound in congenital infections, *Ultrasound Obstet. Gynecol.* (2020).
- [3] J. Born, N. Wiedemann, M. Cossio, C. Buhre, G. Brändle, K. Leidermann, J. Goulet, A. Aujayeb, M. Moor, B. Rieck, et al., Accelerating detection of lung pathologies with explainable ultrasound image analysis, *Appl. Sci.* 11 (2) (2021) 672.
- [4] B. Maranin, G. Ciano, S. Mandrioli, M. Galiè, M. Govoni, The role of ultrasound in temporomandibular joint disorders: An update and future perspectives, *Front. Med.* 9 (2022) 926573.
- [5] B.G. Coleman, J.E. Langer, S.C. Horii, The diagnostic features of spina bifida: the role of ultrasound, *Fetal diagnosis and therapy* 37 (3) (2015) 179–196.
- [6] L. Edwards, L. Hui, First and second trimester screening for fetal structural anomalies, *Semin. Fetal Neonatal Med.* 23 (2) (2018) 102–111, <http://dx.doi.org/10.1016/j.siny.2017.11.005>, URL: <https://www.sciencedirect.com/science/article/pii/S1744165X17301361>, PRENATAL SCREENING.
- [7] J. Karim, N. Roberts, L. Salomon, A. Papageorgiou, Systematic review of first-trimester ultrasound screening for detection of fetal structural anomalies and factors that affect screening performance, *Obstet. Gynecol. Surv.* 73 (4) (2018) 185–187.
- [8] B. Pu, K. Li, S. Li, N. Zhu, Automatic fetal ultrasound standard plane recognition based on deep learning and IoT, *IEEE Trans. Ind. Inform.* 17 (11) (2021) 7771–7780.
- [9] J. Guo, G. Tan, F. Wu, H. Wen, K. Li, Fetal ultrasound standard plane detection with coarse-to-fine multi-task learning, *IEEE J. Biomed. Health Inf.* 27 (10) (2023) 5023–5031, <http://dx.doi.org/10.1109/JBHI.2022.3209589>.
- [10] Q. Huang, H. Tian, L. Jia, Z. Li, Z. Zhou, A review of deep learning segmentation methods for carotid artery ultrasound images, *Neurocomputing* (2023) 126298.
- [11] J. He, L. Yang, B. Liang, S. Li, C. Xu, Fetal cardiac ultrasound standard section detection model based on multitask learning and mixed attention mechanism, *Neurocomputing* (2024) 127443.
- [12] R.A. Filly, V.A. Feldstein, *Ultrasound evaluation of normal fetal anatomy, Ultrasonography in Obstetrics and Gynecology*, fifth ed., Elsevier, Amsterdam, 2007, pp. 297–362.
- [13] Z. Kivilevitch, L. Gindes, H. Deutsch, R. Achiron, In-utero evaluation of the fetal umbilical-portal venous system: two- and three-dimensional ultrasonic study, *Ultrasound Obstet. Gynecol.* 34 (6) (2009) 634–642, <http://dx.doi.org/10.1002/uog.7459>, arXiv:<https://obgyn.onlinelibrary.wiley.com/doi/pdf/10.1002/uog.7459>, URL: <https://obgyn.onlinelibrary.wiley.com/doi/abs/10.1002/uog.7459>.
- [14] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., An image is worth 16x16 words: Transformers for image recognition at scale, 2020, arXiv preprint arXiv:2010.11929.
- [15] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin transformer: Hierarchical vision transformer using shifted windows, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 10012–10022.
- [16] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, S. Zagoruyko, End-to-end object detection with transformers, in: *European Conference on Computer Vision*, Springer, 2020, pp. 213–229.
- [17] F. Li, P. Li, X. Wu, P. Zeng, G. Lyu, Y. Fan, P. Liu, H. Song, Z. Liu, FHUSP-NET: A multi-task model for fetal heart ultrasound standard plane recognition and key anatomical structures detection, *Comput. Biol. Med.* 168 (2024) 107741.
- [18] T. Yang, L. Yuan, P. Li, P. Liu, Real-time automatic assisted detection of uterine fibroid in ultrasound images using a deep learning detector, *Ultrasound Med. Biol.* 49 (7) (2023) 1616–1626.
- [19] A.I. Sapitri, S. Nurmaini, M.N. Rachmatullah, B. Tutuko, A. Darmawahyuni, F. Firdaus, D.P. Rini, A. Islami, Deep learning-based real time detection for cardiac objects with fetal ultrasound video, *Inform. Med. Unlocked* 36 (2023) 101150.
- [20] C. Zhen, H. Wang, J. Cheng, X. Yang, C. Chen, X. Hu, Y. Zhang, Y. Cao, D. Ni, W. Huang, et al., Locating multiple standard planes in first-trimester ultrasound videos via the detection and scoring of key anatomical structures, *Ultrasound Med. Biol.* 49 (9) (2023) 2006–2016.
- [21] H. Wu, B. Wu, F. Lai, P. Liu, G. Lyu, S. He, J. Dai, et al., Application of artificial intelligence in anatomical structure recognition of standard section of fetal heart, *Comput. Math. Methods Med.* 2023 (2023).
- [22] M. Komatsu, A. Sakai, R. Komatsu, R. Matsuoka, S. Yasutomi, K. Shozu, A. Dozen, H. Machino, H. Hidaka, T. Arakaki, et al., Detection of cardiac structural abnormalities in fetal ultrasound videos using deep learning, *Appl. Sci.* 11 (1) (2021) 371.

- [23] Z. Lin, S. Li, D. Ni, Y. Liao, H. Wen, J. Du, S. Chen, T. Wang, B. Lei, Multi-task learning for quality assessment of fetal head ultrasound images, *Med. Image Anal.* 58 (2019) 101548.
- [24] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, D. Krishnan, Supervised contrastive learning, *Adv. Neural Inf. Process. Syst.* 33 (2020) 18661–18673.
- [25] B. Sun, B. Li, S. Cai, Y. Yuan, C. Zhang, Fscf: Few-shot object detection via contrastive proposal encoding, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 7352–7362.
- [26] M. Ghayekhloo, A. Nickabadi, Supervised contrastive learning for graph representation enhancement, *Neurocomputing* 588 (2024) 127710.
- [27] Y. Chen, C. Zhang, L. Liu, C. Feng, C. Dong, Y. Luo, X. Wan, USCL: pretraining deep ultrasound image diagnosis model through video contrastive representation learning, in: *Medical Image Computing and Computer Assisted Intervention—MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part VIII* 24, Springer, 2021, pp. 627–637.
- [28] C. Zhang, Y. Chen, L. Liu, Q. Liu, X. Zhou, Hico: Hierarchical contrastive learning for ultrasound video model pretraining, in: *Proceedings of the Asian Conference on Computer Vision*, 2022, pp. 229–246.
- [29] J. Sun, B. Wu, T. Zhao, L. Gao, K. Xie, T. Lin, J. Sui, X. Li, X. Wu, X. Ni, Classification for thyroid nodule using ViT with contrastive learning in ultrasound images, *Comput. Biol. Med.* 152 (2023) 106444, <http://dx.doi.org/10.1016/j.combiomed.2022.106444>, URL: <https://www.sciencedirect.com/science/article/pii/S0010482522011520>.
- [30] A. Chartsias, S. Gao, A. Mumith, J. Oliveira, K. Bhatia, B. Kainz, A. Beqiri, Contrastive learning for view classification of echocardiograms, in: *Simplifying Medical Ultrasound: Second International Workshop, ASMUS 2021, Held in Conjunction with MICCAI 2021, Strasbourg, France, September 27, 2021, Proceedings 2*, Springer, 2021, pp. 149–158.
- [31] Z. Fu, J. Jiao, R. Yasrab, L. Drukker, A.T. Papageorgiou, J.A. Noble, Anatomy-aware contrastive representation learning for fetal ultrasound, in: *European Conference on Computer Vision*, Springer, 2022, pp. 422–436.
- [32] Z.-M. Chen, X.-S. Wei, P. Wang, Y. Guo, Multi-label image recognition with graph convolutional networks, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 5177–5186.
- [33] W. Ju, Z. Fang, Y. Gu, Z. Liu, Q. Long, Z. Qiao, Y. Qin, J. Shen, F. Sun, Z. Xiao, et al., A comprehensive survey on deep graph representation learning, *Neural Netw.* (2024) 106207.
- [34] W. Ju, Y. Wang, Y. Qin, Z. Mao, Z. Xiao, J. Luo, J. Yang, Y. Gu, D. Wang, Q. Long, et al., Towards graph contrastive learning: A survey and beyond, 2024, arXiv preprint [arXiv:2405.11868](https://arxiv.org/abs/2405.11868).
- [35] W. Ju, S. Yi, Y. Wang, Z. Xiao, Z. Mao, H. Li, Y. Gu, Y. Qin, N. Yin, S. Wang, et al., A survey of graph neural networks in real world: Imbalance, noise, privacy and ood challenges, 2024, arXiv preprint [arXiv:2403.04468](https://arxiv.org/abs/2403.04468).
- [36] W. Ju, Y. Qin, S. Yi, Z. Mao, K. Zheng, L. Liu, X. Luo, M. Zhang, Zero-shot node classification with graph contrastive embedding network, *Trans. Mach. Learn. Res.* (2023).
- [37] J. Chen, S. Hou, L. Feng, B. Lu, M. Yang, F. Sun, Q. Li, T. Tan, X. Deng, G. Wei, Ultrasonic evaluation of fetal lung development using deep learning with graph, *Displays* 78 (2023) 102451.
- [38] J. Yuan, X. Luo, Y. Qin, Z. Mao, W. Ju, M. Zhang, Alex: Towards effective graph transfer learning with noisy labels, in: *Proceedings of the 31st ACM International Conference on Multimedia*, 2023, pp. 3647–3656.
- [39] S. Yi, W. Ju, Y. Qin, X. Luo, L. Liu, Y. Zhou, M. Zhang, Redundancy-free self-supervised relational learning for graph clustering, *IEEE Trans. Neural Netw. Learn. Syst.* (2023).
- [40] X. Luo, W. Ju, Y. Gu, Z. Mao, L. Liu, Y. Yuan, M. Zhang, Self-supervised graph-level representation learning with adversarial contrastive learning, *ACM Trans. Knowl. Discov. Data* 18 (2) (2023) 1–23.
- [41] S. Yin, Q. Peng, H. Li, Z. Zhang, X. You, H. Liu, K. Fischer, S.L. Furth, G.E. Tasian, Y. Fan, Multi-instance deep learning with graph convolutional neural networks for diagnosis of kidney diseases using ultrasound imaging, in: *Uncertainty for Safe Utilization of Machine Learning in Medical Imaging and Clinical Image-Based Procedures: First International Workshop, UNSURE 2019, and 8th International Workshop, CLIP 2019, Held in Conjunction with MICCAI 2019, Shenzhen, China, October 17, 2019, Proceedings 8*, Springer, 2019, pp. 146–154.
- [42] L. Zhao, K. Li, B. Pu, J. Chen, S. Li, X. Liao, An ultrasound standard plane detection model of fetal head based on multi-task learning and hybrid knowledge graph, *Future Gener. Comput. Syst.* 135 (2022) 234–243.
- [43] X. Wang, W. Wang, X. Cai, Automatic measurement of fetal head circumference using a novel GCN-assisted deep convolutional network, *Comput. Biol. Med.* 145 (2022) 105515.
- [44] X. Glorot, A. Bordes, Y. Bengio, Deep sparse rectifier neural networks, in: *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics, JMLR Workshop and Conference Proceedings*, 2011, pp. 315–323.
- [45] M. Gutmann, A. Hyvärinen, Noise-contrastive estimation: A new estimation principle for unnormalized statistical models, in: *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics, JMLR Workshop and Conference Proceedings*, 2010, pp. 297–304.
- [46] Z. Cai, N. Vasconcelos, Cascade r-cnn: Delving into high quality object detection, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 6154–6162.
- [47] P. Sun, R. Zhang, Y. Jiang, T. Kong, C. Xu, W. Zhan, M. Tomizuka, L. Li, Z. Yuan, C. Wang, et al., Sparse r-cnn: End-to-end object detection with learnable proposals, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 14454–14463.
- [48] S. Chen, P. Sun, Y. Song, P. Luo, Diffusiondet: Diffusion model for object detection, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 19830–19843.
- [49] C. Jiang, H. Xu, X. Liang, L. Lin, Hybrid knowledge routed modules for large-scale object detection, *Adv. Neural Inf. Process. Syst.* 31 (2018).
- [50] S. Ren, K. He, R. Girshick, J. Sun, Faster r-cnn: Towards real-time object detection with region proposal networks, *Adv. Neural Inf. Process. Syst.* 28 (2015).
- [51] L. Van der Maaten, G. Hinton, Visualizing data using t-sne., *J. Mach. Learn. Res.* 9 (11) (2008).
- [52] W. Zhang, Z. Sheng, Z. Yin, Y. Jiang, Y. Xia, J. Gao, Z. Yang, B. Cui, Model degradation hinders deep graph neural networks, in: *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2022, pp. 2493–2503.



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